

Software Architecture Design — E-Trade Management System

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1. System Summary

The system is a Web-enterprise e-trade (ET) management platform that supports electronic trading by combining customer-facing services (ordering, product browsing, personalization, payment) with enterprise-side management functions (product/assortment management, marketing, financial processing, resource management, and delivery coordination). It mediates interactions among consumers, customers, the marketing department, store managers, suppliers, and accounting/banking entities, while integrating with external suppliers, manufacturers, accounting systems, banks, and payment systems.

2. Assumptions

The SRS states no assumptions were introduced during extraction; however, some architectural inferences require clarifying assumptions due to ambiguity in the source SRS:

- **A-01:** "Shop window" refers to a customer-facing product display/catalog view generated from assortment and business-rule data (FR03), distinct from the underlying product information store (FR04, FR05).
 - **A-02:** "Unified registration and authorization across showcases" (FR09) implies multiple customer-facing storefronts/showcases share a common identity mechanism, requiring a logically centralized identity-handling capability.
 - **A-03:** "Third-party payment systems" and "accounting, banks" (FR10, FR13) are treated as external systems accessed through defined interfaces, since no protocol or product is specified.
 - **A-04:** C-02 ("each information system handles its own data class") is interpreted as a mandate for decentralized data ownership per component, motivating a modular/distributed component structure rather than a single shared data store.
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3. Selected Architecture Style

Architecture Style

Service-Oriented Modular Architecture (a component-based architecture composed of loosely coupled, independently responsible services communicating through defined interfaces).

Justification

- FR01–FR15 collectively describe distinct, largely independent functional areas (ordering, product info, marketing, customer management, finance, resource management, delivery), each best encapsulated as a separate component — favoring modular decomposition over a monolithic design.
 - NFR-03 (interoperability, platform-independent service access) and C-01 (interface independence from application platform) require components to expose access through platform-neutral service interfaces, which is a defining trait of service orientation.
 - NFR01 (scalability) and NFR-05 (informational integration) favor components that can scale and integrate independently rather than as one tightly bound unit.
 - C-02 (each information system handles its own data class) directly supports decentralized, component-owned data rather than a shared central database — reinforcing modular service boundaries.
 - **Inferred Decision:** The specific style (service-oriented modular architecture, rather than, e.g., a strict microservices or SOA-with-ESB pattern) is inferred, since the SRS does not name an architecture style. This is the minimal-commitment interpretation satisfying modularity, interoperability, and independent data ownership requirements without assuming specific technologies.
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4. Architectural Drivers

Functional Drivers

- **User interaction & order intake** (FR01, FR08): drives a dedicated customer-facing interaction/order-entry capability.
- **Order processing & delivery** (FR02, FR12, FR15): drives order management and delivery management components.
- **Shop window generation** (FR03): drives a presentation/catalog-generation component distinct from raw product data management.
- **Product information lifecycle** (FR04, FR05): drives a product information management component.
- **Marketing/advertising** (FR06): drives a marketing management component.
- **Customer management/personalization** (FR07): drives a customer management component.
- **Unified identity** (FR09): drives a shared registration/authorization capability.
- **Payment interaction** (FR10, FR13): drives an external financial interface component.
- **Market/supplier monitoring** (FR11): drives a supplier/market monitoring component.
- **Resource management** (FR14): drives a Web-enterprise resource management component.

Quality Attribute Drivers

- **Scalability** (NFR01)
- **Flexibility** (NFR-02)
- **Interoperability** (NFR-03)
- **Availability** (NFR-04)
- **Integration** (NFR-05)

Influence on Architecture

Scalability and flexibility drive component independence, so individual functions (e.g., order management, marketing) can evolve or scale without affecting others. Interoperability and the platform-independence constraint (C-01) drive the use of well-defined service interfaces between components rather than tight, platform-specific coupling. Availability drives an architecture where customer-facing components (interaction, shop window, ordering) are structured for continuous online operation. Integration (NFR-05) drives explicit, well-defined interaction paths between internal components and external systems (suppliers, accounting, banks, payment systems).

5. Requirements-to-Architecture Traceability

Requirement ID	Requirement	Architectural Component(s)	Justification
FR01	Interact with users, receive orders, collect primary information	User Interaction Component	Directly implements user-facing intake described in FR01
FR02	Process orders, manage delivery, support business processes	Order Management Component, Delivery Management Component	Order processing and delivery are explicitly named
FR03	Generate shop window data from assortment and business rules	Shop Window Management Component	Directly matches requirement wording
FR04	Provide detailed product information	Product Information Management Component	Direct match
FR05	Gather, structure, update, process product information	Product Information Management Component	Same component owns full product data lifecycle
FR06	Manage advertising and marketing	Marketing & Advertising Component	Direct match
FR07	Support customer management and personalization	Customer Management Component	Direct match
FR08	Identify consumers, support order creation	User Interaction Component, Order Management Component	Identification precedes order creation, spanning both
FR09	Unified registration/authorization across showcases	Identity & Access Component	Centralization implied by "unified" (A-02)
FR10	Interact with third-party payment systems	Financial Management Component	Payment interaction is a financial concern
FR11	Monitor markets, suppliers, prices	Supplier & Market Monitoring Component	Direct match

FR12	Verify orders, monitor execution	Order Management Component	Order verification/monitoring is core order logic
FR13	Interact with accounting, banks, payment systems	Financial Management Component	Direct match
FR14	Manage Web-enterprise resources	Resource Management Component	Direct match
FR15	Manage order delivery	Delivery Management Component	Direct match
NFR01	Scalability	All components (via modular decomposition)	Independent components enable independent scaling
NFR-02	Flexibility	Service Interface layer	Loosely coupled interfaces allow adaptable reconfiguration
NFR-03	Interoperability	Service Interface layer	Platform-independent access per C-01
NFR-04	Availability	User Interaction, Shop Window, Order Management	Customer-facing components require continuous availability
NFR-05	Integration	Integration/External Interface Component	Coordinates cross-component and external data integration
C-01	Platform-independent interface	Service Interface layer	Directly implements constraint
C-02	Component-owned data classes	All components (data ownership model)	Directly implements constraint

6. Major Architectural Components

1. User Interaction Component

- **Responsibility:** Receive user input, collect primary order/customer information (FR01, FR08).
- **Inputs:** Consumer/customer requests, identification data.
- **Outputs:** Structured order intake data, user session context.
- **Interfaces:** Service interface to Order Management, Identity & Access.
- **Dependencies:** Identity & Access Component.

2. Identity & Access Component

- **Responsibility:** Unified registration and authorization across showcases (FR09).
- **Inputs:** Registration/authentication requests.
- **Outputs:** Authenticated identity/session tokens.
- **Interfaces:** Service interface exposed to all customer-facing components.
- **Dependencies:** None internal; foundational service.

3. Shop Window Management Component

- **Responsibility:** Generate shop window (catalog display) data from product assortment and business rules (FR03).
- **Inputs:** Product data, business rules.
- **Outputs:** Presentable shop window/catalog data.
- **Interfaces:** Consumes Product Information Management output; serves User Interaction.
- **Dependencies:** Product Information Management Component.

4. Product Information Management Component

- **Responsibility:** Gather, structure, update, and provide detailed product information (FR04, FR05).
- **Inputs:** Supplier/manufacturer product data.
- **Outputs:** Structured, current product records.
- **Interfaces:** Serves Shop Window Management, Supplier & Market Monitoring.
- **Dependencies:** Supplier & Market Monitoring Component.

5. Order Management Component

- **Responsibility:** Process orders, verify orders, monitor execution (FR02, FR08, FR12).
- **Inputs:** Order intake data, identity context.
- **Outputs:** Verified orders, order status.
- **Interfaces:** Interacts with Delivery Management, Financial Management, User Interaction.
- **Dependencies:** User Interaction, Identity & Access, Financial Management.

6. Delivery Management Component

- **Responsibility:** Manage order delivery (FR02, FR15).
- **Inputs:** Verified order data.
- **Outputs:** Delivery status/tracking data.
- **Interfaces:** Consumes Order Management output.
- **Dependencies:** Order Management Component.

7. Marketing & Advertising Component

- **Responsibility:** Manage advertising and marketing activities (FR06).
- **Inputs:** Product/assortment data, campaign parameters.
- **Outputs:** Marketing content/campaign data.
- **Interfaces:** Supplies data to Shop Window Management.
- **Dependencies:** Product Information Management Component.

8. Customer Management Component

- **Responsibility:** Support customer management and personalization (FR07).
- **Inputs:** Customer profile/interaction data.
- **Outputs:** Personalization parameters, customer records.
- **Interfaces:** Interacts with User Interaction, Marketing & Advertising.
- **Dependencies:** Identity & Access Component.

9. Financial Management Component

- **Responsibility:** Interact with accounting, banks, and payment systems (FR10, FR13).

- **Inputs:** Order/payment data.
- **Outputs:** Payment confirmations, financial records.
- **Interfaces:** External interface to banks/payment systems; internal interface to Order Management.
- **Dependencies:** External Integration Component.

10. Supplier & Market Monitoring Component

- **Responsibility:** Monitor markets, suppliers, and prices (FR11).
- **Inputs:** External supplier/market data.
- **Outputs:** Updated pricing/supplier data.
- **Interfaces:** Supplies Product Information Management; external interface to suppliers/manufacturers.
- **Dependencies:** External Integration Component.

11. Resource Management Component

- **Responsibility:** Manage Web-enterprise resources (FR14).
- **Inputs:** Resource utilization/configuration data.
- **Outputs:** Resource allocation status.
- **Interfaces:** Cross-cutting service consumed by other internal components.
- **Dependencies:** None specified beyond internal components.

12. External Integration Component

- **Responsibility:** Manage interaction with external entities — suppliers, manufacturers, accounting, banks, payment systems (Dependencies §2.2, FR10, FR13).
- **Inputs/Outputs:** External data exchange in both directions.
- **Interfaces:** Platform-independent service interfaces (C-01) to external systems.
- **Dependencies:** Financial Management, Supplier & Market Monitoring.

7. Component Interactions

Data Flow: Product data flows from External Integration (suppliers/manufacturers) → Product Information Management → Shop Window Management → User Interaction (customer-facing display). Order data flows from User Interaction → Order Management → Delivery Management and Financial Management. Financial and delivery status data flow back toward Order Management for monitoring (FR12).

Control Flow: User Interaction initiates order-related control flow after authorization by Identity & Access. Order Management coordinates downstream control across Delivery Management and Financial Management. Marketing & Advertising and Customer Management operate on parallel, loosely coupled control paths feeding into Shop Window Management and User Interaction respectively.

Communication Mechanisms: All inter-component communication occurs through defined service interfaces to satisfy platform independence (C-01) and interoperability (NFR-03). The SRS does not specify a communication protocol; therefore, the specific mechanism

(synchronous request/response vs. asynchronous messaging) is **not assumed** and is left as an implementation-level decision.

8. Data Management Strategy

Data Ownership

Per C-02 ("each information system handles its own data class"), each component logically owns its respective data:

- Product Information Management → product data
- Order Management → order data
- Customer Management → customer/personalization data
- Financial Management → financial/payment records
- Delivery Management → delivery/tracking data
- Supplier & Market Monitoring → supplier/market/pricing data

Persistence Strategy

Each component persists its own data class independently (per C-02), without a shared central store. No specific persistence technology is assumed, as none is specified in the SRS.

Data Flow

Data originates externally (suppliers, manufacturers, banks) or from user interaction, flows into the owning component, and is exposed to consuming components strictly through service interfaces rather than direct data access — preserving component-level data ownership.

Data Storage Approach

Logical, decentralized storage aligned to component boundaries (one logical data store per owning component), consistent with C-02. No physical database technology is specified or assumed.

9. External System Interactions

- **Suppliers:** Provide product/assortment and market data (FR11, §2.2).
- **Manufacturers:** Provide product data (§2.2).
- **Accounting:** Financial data exchange (FR13).
- **Banks:** Payment/financial settlement (FR13).
- **Payment Systems:** Third-party transaction processing (FR10).

All external interactions are mediated through the External Integration Component via platform-independent interfaces (C-01).

10. Quality Attribute Considerations

- **Functional Coverage:** All FR01–FR15 are mapped to a dedicated component (§5), ensuring complete functional coverage.
 - **Modularity:** Each business function (ordering, product info, marketing, finance, delivery) is encapsulated in an independent component, directly supporting C-02 and NFR-02.
 - **Maintainability:** Component isolation allows updates to one functional area (e.g., Marketing & Advertising) without impacting others.
 - **Scalability:** Independent components can scale individually based on load (NFR01), e.g., Order Management under peak order volume.
 - **Interoperability:** Platform-independent service interfaces (C-01) allow heterogeneous client platforms to access services (NFR-03).
 - **Availability:** Customer-facing components (User Interaction, Shop Window Management) are structured for continuous online operation (NFR-04).
 - **Reliability:** Order verification and monitoring (FR12) provide a checkpoint for consistent order state, supporting reliable order processing.
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11. Architectural Decisions and Trade-offs

Decision	Reason	Advantages	Potential Disadvantages
Adopt service-oriented modular architecture	NFR-03, C-01 require platform-independent, interoperable access	Interoperability, flexibility, independent scalability	Added interface/integration overhead
Decentralized, component-owned data (no shared database)	C-02 explicitly mandates per-system data ownership	Clear ownership boundaries, supports modularity	Requires careful cross-component data consistency management
Separate Shop Window Management from Product Information Management (Inferred)	FR03 describes shop window generation as distinct from FR04/FR05 product data management	Presentation logic decoupled from core product data	Introduces an additional integration point
Centralize Identity & Access as shared service (Inferred)	FR09 requires "unified" registration/authorization across showcases	Consistent identity across all customer touchpoints	Single logical dependency point for authentication
Dedicated External Integration Component (Inferred)	Multiple external dependencies (§2.2, FR10, FR13) share the need for platform-independent access (C-01)	Centralizes external interface management, easier to maintain	Potential bottleneck if not internally decomposed further

No communication protocol assumed	SRS does not specify protocol	Remains technology-neutral, avoids unwarranted assumptions	Leaves interaction mechanism underspecified for implementation
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12. Architectural Risks and Limitations

Potential Architectural Risks

- Absence of specified communication protocols could lead to inconsistent implementation choices across components.
- Financial Management and External Integration components handling sensitive data (payments, banking) lack explicit security requirements in the SRS, posing a risk if security is not addressed during implementation.

Architectural Limitations

- The SRS provides no explicit non-functional requirement for security, despite handling financial and personal customer data this is a limitation of the source requirements, not the architecture.
- No explicit performance or response-time requirements are stated, limiting the ability to architect for specific performance targets.

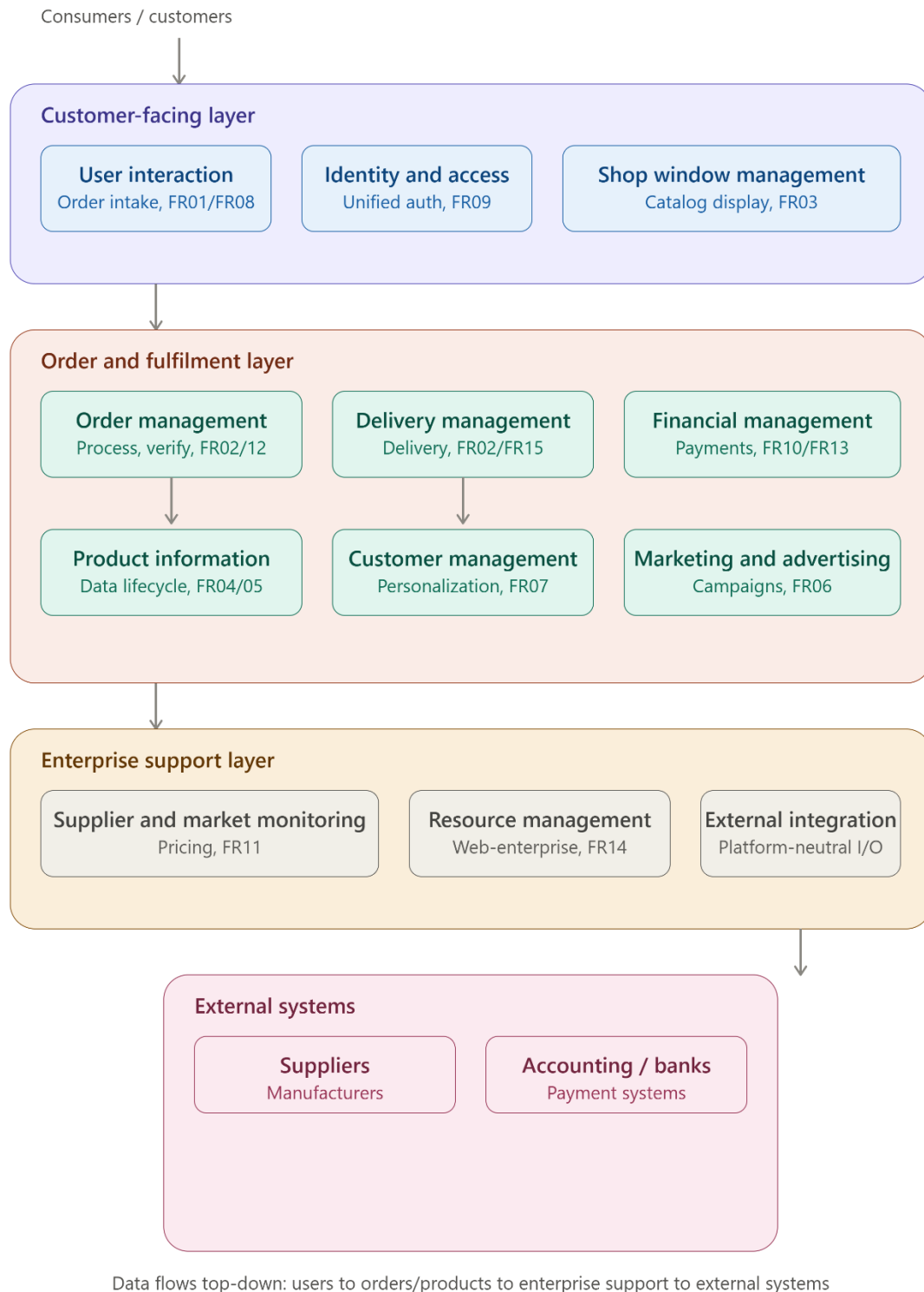
Areas Requiring Future Refinement

- Definition of specific interface contracts between components (currently described only functionally).
- Clarification of data consistency mechanisms across independently owned data classes (per C-02).
- Explicit security and authentication detail beyond "unified registration and authorization" (FR09).

(Limitations above are due to incomplete SRS content, not inherent flaws in the proposed architecture.)

13. Mermaid Architecture Model

Architecture Model Diagram



Architecture Description

- **Customer-facing layer** (blue): User Interaction handles order intake and initial user contact (FR01, FR08); Identity & Access provides unified registration/authorization across showcases (FR09); Shop Window Management generates the catalog view from product and marketing data (FR03).

- **Order and fulfilment layer** (teal): Order Management processes and verifies orders (FR02, FR12), feeding Delivery Management (FR15) and Financial Management (FR10, FR13). Product Information, Customer Management, and Marketing & Advertising supply the data these depend on (FR04–FR07).
 - **Enterprise support layer** (gray): Supplier & Market Monitoring (FR11) and Resource Management (FR14) support internal operations, while External Integration provides the single, platform-independent gateway to outside systems (C-01).
 - **External systems** (pink): Suppliers/manufacturers and accounting/banks/payment systems, accessed only through External Integration.
 - **Data flow**: top-down — consumer requests enter through the customer-facing layer, are processed and fulfilled in the order layer, drawing on enterprise support data, which in turn synchronizes with external systems.
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14. Conclusion

The proposed service-oriented modular architecture satisfies all fifteen functional requirements (FR01–FR15) by assigning each a clearly bounded, traceable component, avoiding overlap or unaddressed functionality. Architecturally significant quality attributes are directly supported: scalability (NFR01) and flexibility (NFR-02) through independently deployable, loosely coupled components; interoperability (NFR-03) and the platform-independence constraint (C-01) through uniform service interfaces; availability (NFR-04) through dedicated, continuously operable customer-facing components; and informational integration (NFR-05) through an explicit External Integration Component mediating all cross-boundary data exchange. The component-owned data model directly satisfies constraint C-02. The selected style is appropriate because the SRS describes a system composed of largely independent business capabilities (ordering, product management, marketing, finance, delivery) serving multiple distinct user types, which is best served by modular, interface-driven decomposition rather than a monolithic or tightly centralized design — while introducing no technology, protocol, or pattern beyond what the SRS requires or reasonably implies.